



- Passive Components, Transformer, Independent & dependent sources, KCL, KVL
- Electronic devices: *pn* junction diode, Zener diode, LED, Photodiode, solar cell
- Diode circuits – half-wave and bridge rectifiers
- Operational amplifiers, feedback circuits, and comparators
- Logic gates and digital circuits, digital-to-analog converter (DAC), and analog-to-digital converter (ADC)
- Microprocessors, microcontrollers, memory and I/O devices
- Microcontroller board (Arduino) with real-world interfacing
- BJT and MOSFET switches
- Electromechanical devices: Relays, DC motors, Servo motors

- Two-terminal (Single-port) devices
 - diodes (and special types of diodes)
- Three-terminal (Two-port) active devices
 - Bipolar Junction Transistors (BJT), and
 - Metal-oxide Semiconductor Field-effect Transistors (MOSFET)
- Passive circuits
 - Made of R, L, and C. Eg. Filters (Low-pass filter, high-pass filter, band-pass filter)
- Active circuits (made using BJTs/MOSFETs, and passive elements R, L and C)
 - Amplifiers
 - Voltage Regulators
 - Sinusoidal oscillators
 - Waveform generators

DC - 0 freq

RC High-Pass Filter

- Circuit diagram of a high-pass filter is shown.

- V_{out}/V_{in} in terms of impedances of R and C:
 $V_{out}/V_{in} = R/[R + (1/j\omega C)] = 1/[1 + 1/(j\omega CR)]$
 where $\omega = 2\pi f$

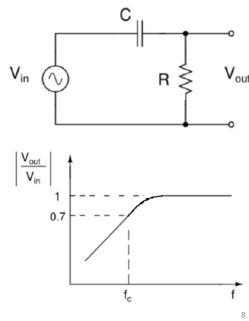
- Gain = magnitude of (V_{out}/V_{in}) :

$$A = \left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{[1 + (\omega CR)^2]^{1/2}}$$

$$= \frac{1}{[1 + (\omega_c/\omega)^2]^{1/2}} = \frac{1}{[1 + (f_c/f)^2]^{1/2}}$$

- where $\omega_c = 1/(RC)$ and $f_c = 1/(2\pi RC)$.

- The sketch shows $|V_{out}/V_{in}|$ as a function of f .
 At $f = f_c$, $A = 1/2^{1/2} \approx 0.707$.



MS101-EE-Lect-1

Dependent Sources

- Four types as shown
- In Fig 4(a) and (c), K is a dimensionless scaling constant.
- In Fig 4(b), g is a scaling factor with units of A/V.
- Fig. 4(d), r is a scaling factor with units of V/A.
- The controlling current i_x and the controlling voltage v_x must be defined in the circuit.

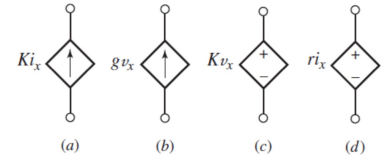


Fig. 4 Circuit symbols for the four different types of dependent sources:
 (a) current-controlled current source;
 (b) voltage-controlled current source;
 (c) voltage-controlled voltage source;
 (d) Current controlled voltage source

RC Low-Pass Filter

- Circuit diagram of a Low-pass filter is shown.

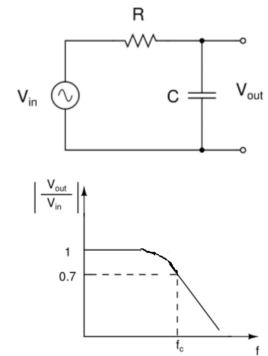
- V_{out}/V_{in} in terms of impedances of R and C:
 $V_{out}/V_{in} = (1/j\omega C) / [R + (1/j\omega C)] = 1/[1 + j\omega CR]$,
 where $\omega = 2\pi f$
- Gain = magnitude of (V_{out}/V_{in}) :

$$A = \left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{[1 + (\omega CR)^2]^{1/2}}$$

$$= \frac{1}{[1 + (\omega/\omega_c)^2]^{1/2}} = \frac{1}{[1 + (f/f_c)^2]^{1/2}}$$

- where $\omega_c = 1/(RC)$ and $f_c = 1/(2\pi RC)$.

- The sketch shows $|V_{out}/V_{in}|$ as a function of f .
 At $f = f_c$, $A = 1/2^{1/2} \approx 0.707$.



The *pn* junction Terminal Characteristic

- The terminal characteristic of a *pn* junction diode (i.e. i vs v) can be written as:

$$i = I_s [\exp(v/V_T) - 1],$$

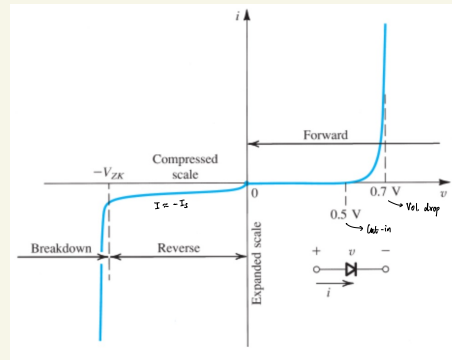
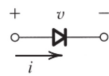
- where I_s = reverse saturation current (typ of the order of 10^{-15} A)

- $V_T = (kT/q)$ is the thermal voltage ($V_T \approx 25$ mV at 20°C)

- k = Boltzmann's constant = 8.62×10^{-5} eV/K = 1.38×10^{-23} joules/kelvin

- T = the absolute temperature in kelvins = $273 + \text{temperature in } ^\circ\text{C}$

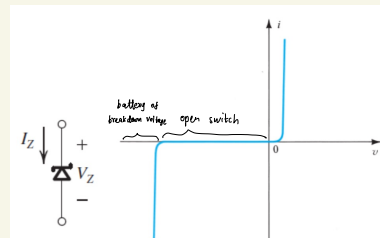
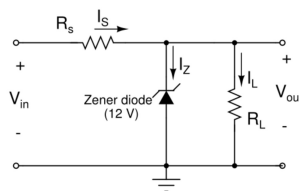
- q = the magnitude of electronic charge = 1.60×10^{-19} coulomb



Diode breakdown is not destructive

- Zener regulator circuit

- Output voltage V_{out} will be reasonably constant for
 - Variations in V_{in} and
 - Variations in I_L



D. Light Emitting Diodes

- LEDs are special diodes that emit light when **forward biased** (i.e. they convert electrical energy into light). They are called electrical-to-optical converters.
- Light is produced due to *radiative recombination* of injected minority carriers with the majority carriers.
- For radiative recombination, the *pn* junction should be made using a semiconductor of the type known as *direct-bandgap materials*.
- Examples of direct-bandgap material
 - Gallium arsenide (GaAs), AlGaAs, GaN, InP, ..
 - Wavelength of an LED's light depends on the electronic bandgap of the material used
- The light emitted by an LED is proportional to the number of recombinations that take place, i.e. *proportional to the forward current in the diode*.

E. Photodiodes

- When a *pn* junction is **reverse-biased**, a depletion region is formed around the *pn* junction.
 - Only the reverse saturation current flows, which is very small and is primarily due to the minority carriers.
- If the reverse-biased *pn* junction is exposed to incident light
 - the photons impacting the junction cause covalent bonds to break, and thus electron-hole pairs are generated in the depletion layer.
 - The electric field in the depletion region then sweeps the liberated electrons to the *n* side and the holes to the *p* side, giving rise to a reverse current across the junction.
 - This current is known as **photocurrent**, and is proportional to the intensity of the incident light.
 - Such a diode is called a **photodiode**
 - It can be used to convert light signals into electrical signals.

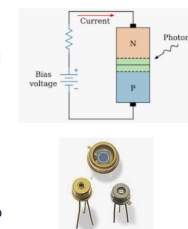


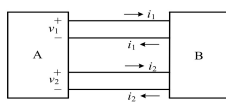
Fig. 11
 Source: Electric Circuit Studio

Differential signals: Each signal needs two conductors.

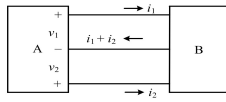
Single-ended signals: Several signals share a common reference in the case of voltage signals, & a common return in the case of current signals.

Circuit ground: A conductor or terminal (usually attached to a power supply terminal) serving as the common reference for several voltages or the common return path for several currents.

Grounded signals: Single-ended signals with circuit ground as the reference. These signals are preferred over differential signals, as they need fewer conductors and require simpler circuits. Ground interconnection is usually not explicitly shown in circuit diagrams.



Two differential signals (each with two conductor)



Two single-ended signals (common reference & return)

Amplifier: A circuit or device for increasing the power of the input signal(s) using the power from dc source(s).

Four types of single-ended amplifiers modeled as dependent sources

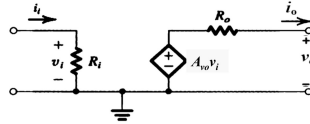
i) Voltage amplifier

(voltage input, voltage output)

Model: Voltage-controlled voltage source (VCVS)

Open-circuit voltage gain: A_{vo}

Ideal VCVS: $R_i = \infty$, $R_o = 0$



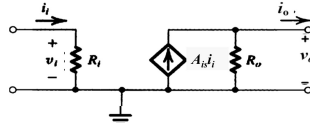
ii) Current amplifier

(current input, current output)

Model: Current-controlled current source (CCCS)

Short-circuit current gain: A_{is}

Ideal CCCS: $R_i = 0$, $R_o = \infty$



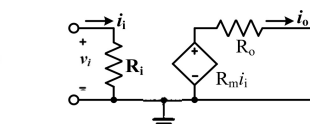
iii) Trans-resistance amplifier

(current input, voltage output)

Model: Current-controlled voltage source (CCVS)

Open-circuit trans-resistance: R_m

Ideal CCVS: $R_i = 0$, $R_o = 0$



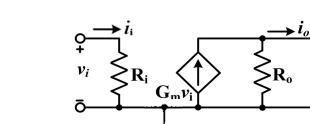
iv) Trans-conductance amplifier

(voltage input, current output)

Model: Voltage-controlled current source (VCCS)

Short-circuit trans-conductance: G_m

Ideal VCCS: $R_i = \infty$, $R_o = \infty$



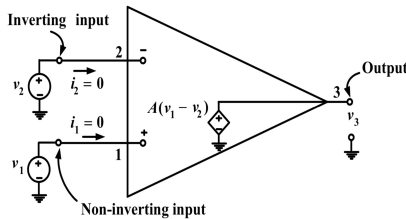
Differential amplifier

- Differential input voltage & single-ended output voltage
- Used to amplify the DM voltage while rejecting the CM voltage.

DM input voltage: $v_{id} = v_1 - v_2$
CM input voltage: $v_{ic} = (v_1 + v_2)/2$
 $v_1 = v_{ic} + v_{id}/2$; $v_2 = v_{ic} - v_{id}/2$

Ideal differential amplifier

- Zero input currents.
- Amplifies the DM input voltage.
- Rejects the CM input voltage.
- Single-ended output voltage: $v_3 = A v_{id}$



Feedback amplifier

- Signal-flow diagram: input x_s , output x_o (voltage or current).

- Amplifier: input x_i , open-loop gain A , output x_o .

- Feedback network: input x_o , feedback factor β , feedback signal x_f .

- Adder: inputs x_s & x_f , output: x_i . $\text{loop gain} = A\beta$

$$x_f = \beta x_o, \quad x_i = x_s - x_f, \quad x_o = Ax_i = A(x_s - \beta x_o) \Rightarrow x_o(1 + A\beta) = Ax_s \Rightarrow x_o = Ax_s / (1 + A\beta).$$

- Feedback amplifier gain (closed-loop gain) $A_f = \frac{x_o}{x_s} = \frac{A}{1 + A\beta}$. For $A\beta \gg 1$, $A_f \approx \frac{1}{\beta}$.

A may have large variability due to electronic device parameters. The circuit can be designed such that β depends only on passive components and thus has a precise value. Example: $A = 10^3$ to 10^5 & $\beta = 1/10$.

(i) $A = 10^3 \Rightarrow A_f = 10^3 / (1 + 10^2) = 10 / (1 + 10^{-2}) = 9.900$. (ii) $A = 10^5 \Rightarrow A_f = 10^5 / (1 + 10^4) = 9.999$.

- For $A\beta \gg 1$, A_f can be precise despite variability in A .

- Negative feedback can be used for obtaining precise gain, but with much less gain.

Input-output relation of an op amp

- Terminal voltages with reference to Gnd: inputs v_1 & v_2 , output v_3 .

- DM input: $v_{id} = v_1 - v_2$.

- CM input: $v_{ic} = (v_1 + v_2)/2$.

- Output voltage $v_3 = A_d v_{id} + A_c v_{ic}$.

- where A_d = DM gain, A_c = CM gain.

- Common-mode rejection ratio:

$$\text{CMRR} = A_d / A_c.$$

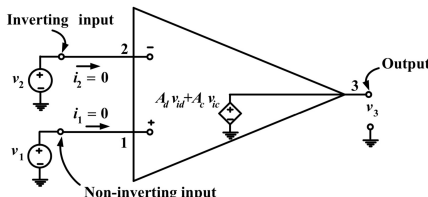
Ideal op amp

- $A_d \rightarrow \infty$, $A_c \rightarrow 0$. $v_3 = A_d(v_1 - v_2)$. $\text{CMRR} = A_d / A_c \rightarrow \infty$.

- DM input is amplified, with no effect of CM input on output. Required DM input $\rightarrow 0$ for finite output.

- Infinite input resistance for both inputs: zero input currents.

- Zero output resistance: output voltage unaffected by the load current.



Logic Reduction . . . contd.

As an example of logic simplification, let us take the function F_1 which we had derived in the sum of products form earlier.

$$F_1 = \bar{A} \cdot \bar{B} \cdot C + \bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C + A \cdot B \cdot C$$

Using the logic identity $x + x = x$, we can write the above function as:
 $F_1 = (\bar{A} \cdot \bar{B} \cdot C + \bar{A} \cdot B \cdot C) + (\bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C) + (\bar{A} \cdot B \cdot C + A \cdot B \cdot C)$
where we have replicated the third term and added it to first and second terms. This can be simplified to

$$\bar{A} \cdot (\bar{B} + B) \cdot C + \bar{A} \cdot B \cdot (\bar{C} + C) + (\bar{A} + A) \cdot B \cdot C$$

Using the identity $x + \bar{x} = 1$, we can write the above as

$$\bar{A} \cdot C \cdot 1 + \bar{A} \cdot B \cdot 1 + 1 \cdot B \cdot C = \bar{A} \cdot C + \bar{A} \cdot B + B \cdot C = \bar{A} \cdot (B + C) + B \cdot C$$

It is easy to verify that this much simpler function evaluates to '1' for ABC combinations of (001), (010), (011) and (111), which agree with the truth table from which we had derived the sum of products function.

Full Adder

(Karnaugh Map)

Full Adder Truth Table:

A	B	C_{in}	Sum	C_{out}
0	0	0	0	0
0	1	0	1	0
1	0	0	1	0
1	1	0	0	1
0	0	1	1	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	1

It gives the following K maps:

AB	00	01	11	10
C_{in} 0	0	1	0	1
C_{in} 1	1	0	1	0

SUM

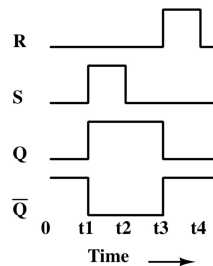
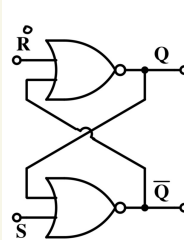
AB	00	01	11	10
C_{in} 0	0	0	1	0
C_{in} 1	0	1	1	1

CARRY

$$\text{sum} = \bar{A} \cdot \bar{B} \cdot C_{in} + \bar{A} \cdot B \cdot \bar{C}_{in} + A \cdot B \cdot C_{in} + A \cdot \bar{B} \cdot \bar{C}_{in} = A \oplus B \oplus C$$

$$C_{out} = A \cdot B + B \cdot C_{in} + C_{in} \cdot A = A \cdot B + C_{in} \cdot (A + B)$$

RS Latch: Set and Reset Action



Initially $R = S = '0'$ and the latch is in reset ($Q = '0'$, $\bar{Q} = '1'$) state.

At t_1 , S goes to '1' and it forces Q to '0' unconditionally. Since $R = '0'$ and $\bar{Q} = '0'$, Q goes to '1'.

At t_2 , S returns to '0', and the latch remains in set ($Q = '1'$, $\bar{Q} = '0'$) state.

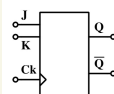
At t_3 , R goes to '1'. It forces Q to '0' unconditionally.

Since $S = '0'$ and $Q = '0'$, $\bar{Q}$ goes to '1'.

at t_4 , R returns to '0'. With $R = 0$, $S = 0$, the latch remains in reset ($Q = '0'$, $\bar{Q} = '1'$) state.

If Set and Reset are simultaneously applied, both Q and $\bar{Q}$ are forced to '0'.

Other Circuits with Memory: JK Flipflop



JK flipflop:

This circuit has two inputs J and K –

These act similarly to S and R inputs in the RS latch.

At the active edge of the clock,

If $J = '0'$ and $K = '0'$, the flipflop outputs remain unchanged.

If $J = '1'$ and $K = '0'$, Q goes to '1' and $\bar{Q}$ goes to '0'.

If $J = '0'$ and $K = '1'$, Q goes to '0' and $\bar{Q}$ goes to '1'.

If $J = '1'$ and $K = '1'$, it complements the values at Q and $\bar{Q}$.

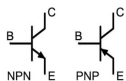
(This and the presence of clock distinguish it from RS latch).

Toggle flipflop: In this, the J and K inputs are tied together and are used as the T input.

When $T = '0'$, the flipflop outputs Q and $\bar{Q}$ remain unchanged.

When $T = '1'$, it toggles ie complements the values at Q and $\bar{Q}$ at the active clock edge.

Circuit symbol



Active mode operation

- BE junction is forward biased.
- BC junction is reversed biased.

Digital and Analog I-O ... contd.

- The function **analogRead** provides the capability of reading the analog voltage on any of the analog input pins and converting it to a 10 bit digital value.
- **analogRead** accepts a channel number as its input and returns a 10 bit integer – which is the converted value from the ADC. For example,
int sensorValue = analogRead(A0);
declares the variable sensorValue to be a (16 bit) integer and places the 10 bit ADC value corresponding to the analog input on channel A0 in it.
- Channels A0 to A5 are available for ADC input.
- The library also has a mapping function which can map this 10 bit range onto a given range – say 0 to 100.

The Arduino library provides many functions which permit one to use the functionality provided by the micro-controller. We have seen many of these already.

Digital I/O: **digitalRead(pin);** returns a **HIGH** or **LOW** value depending on the voltage level on the pin.

digitalWrite(pin, value); writes **HIGH** or **LOW** to a pin.

pinMode(type); sets the direction for I-O on a digital pin. The type can be **INPUT**, **INPUT_PULLUP** or **OUTPUT**.

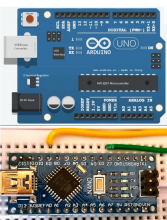
If the type is **INPUT_PULLUP**, an internal pull up circuit is attached to the input pin. This is useful when the external driver is of open collector or open drain type.

Arduino Library Functions for Analog I-O

int analogRead(pin); returns the 10 bit ADC value corresponding to the analog voltage on the given pin.

void analogWrite(pin, PWMvalue); will output a PWM waveform on the given pin with the specified duty cycle (range 0 to 255).

void analogReference (type); configures the reference voltage used for A to D conversion. The argument "type" can be:



DEFAULT – to use the default analog reference of 5 volts (on 5V Arduino boards) or

INTERNAL – to use a built-in reference equal to 1.1 volts (= silicon band gap), or

EXTERNAL – to use the voltage applied to the AREF pin (0 to 5V only) as the reference.

This function must be called **before** analogRead to avoid shorting internal and external references.

Other I/O: **void tone(pin, frequency, [duration]);** A call to this function generates a square wave on the given pin with the specified frequency for the given duration. The third argument is optional. If duration is not given, the output will continue till a call is made to the function:

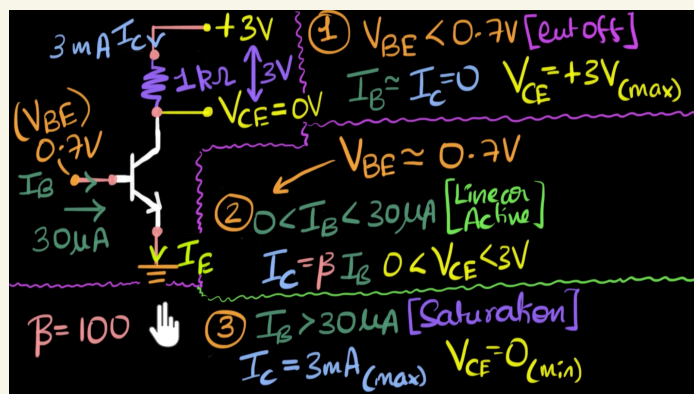
void noTone(pin);

1. Write all the correct options out of the following statements regarding Digital Multimeter (DMM).

- A) The input resistance of a DMM for voltage measurements is typically **10 MΩ**.
 B) It should not be used for measuring AC voltage if the frequency is much higher than 50 Hz or the waveform is non-sinusoidal.
 C) It is set for ohm measurement and connected across a capacitor. The reading will be "1".
 D) It is set for ohm measurement and connected across an inductor. The reading will be "1".
 E) The instrument may be damaged if it is set for ohm measurement and connected across a voltage.

Answer(s): **B, C, D**

Mark: 1



- **Program Counter (PC):** Holds the address of the *next* instruction to be fetched/executed.
- **Data Pointer (DP) or similar (e.g., DPTR in 8051):** Used for data addressing, not for sequencing instructions.

They serve different purposes, so the program counter is *not* the same as the data pointer.

A is incorrect.

B)

"The counter which provides addresses to the program memory is called a program counter."

- This is precisely what the program counter does: it "counts" (or holds) the address in program memory of the current/next instruction.

B is correct.

E)

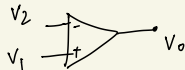
"Moving to a different block of instructions may be accomplished by loading a different value into the program counter."

- The program counter determines *which* instruction address the CPU fetches next. Modifying (loading) the PC directly (or via a jump instruction) changes the flow of execution to a new location.

E is correct.

Superposition $\rightarrow$ for linear : $R, L, C, \diamond$ (dependent)
 $\rightarrow$ of only current & voltage sources.

Op Amp :

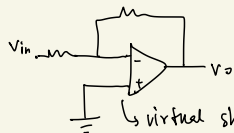


$$V_o = A(V^+ - V^-)$$

Adding feedback helps in stabilising V_o

Linear region of operation $\rightarrow$ Apply -ve feedback, Output voltage $\leq$ supply

Non " " " " $\rightarrow$ " +ve "



virtual short, no input current into op amp

linear iff virtual short

non-linear

also

5.1, 5.2

5.4. Difference Amplifier superposition

It combines inverting and non-inverting amplification for difference amplification.

Select $R_2/R_1 = R_4/R_3 = \alpha$.

Virtual short assumption: $i_1 = i_2$ & $i_3 = i_4$.

Use the superposition method. Find voltage gain & input resistance for each input by setting the other input as 0.

Circuit function: (i) attenuator & non-inverting amplifier for v_1 , (ii) inverting amplifier for v_2 .

$$v_o = \left[\frac{R_2}{R_1 + R_2} v_1 \right] \left[1 + \frac{R_4}{R_3} \right] - \left[\frac{R_4}{R_3} v_2 \right]$$

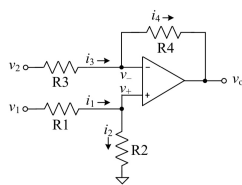
$$= \left[\frac{\alpha}{1 + \alpha} v_1 \right] [1 + \alpha] - [\alpha v_2] = \alpha(v_1 - v_2)$$

• DM gain $A_d = \alpha$. CM gain $A_c = 0$. $R_{id} = R_1 + R_2$. $R_{oD} = R_1$.

• Problems: (i) matched resistances needed, (ii) difficult gain control, (iii) unequal input resistances.

• A voltage v_3 (or DC bias) can be added to the output by connecting R_2 to this voltage in place of ground.

$$v_o = \alpha(v_1 - v_2) + \left[\frac{1}{1 + \alpha} v_3 \right] [1 + \alpha] = \alpha(v_1 - v_2) + v_3$$



+ve feedback $\rightarrow$ Non-linear region

output depends on supply voltage

$$\begin{array}{ll} V_1 > V_2 & +V_s \\ V_1 < V_2 & -V_s \end{array}$$

CMOS:

n-mos: input at gate is high $1 \rightarrow$ ON
 low $0 \rightarrow$ OFF

reverse for p-mos.

Input to TTL unconnected taken as 1